



The crucial role of motorcycle-based ride-hailing among commuters: The case of Jakarta and Bandung metropolitan areas

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ABSTRACT

In Indonesia, issues in the metropolitan transportation sector are considered severe. For instance, peripheral districts in the Jakarta metropolitan area (JMA), such as Bogor, Depok, Tangerang and Bekasi, contribute to the high mobility rate within the metropolitan area. This article examines the extent of motorcycle-based ride-hailing for commuting in the JMA and Bandung metropolitan areas (BMA). In particular, this study explores ride-hailing users' characteristics—social, economic and spatial—and their proportion to the total commuter mobility in a metropolitan area. The study employed descriptive statistical models using commuter data in the JMA and BMA, respectively. Our results indicate that young and lower-income commuters are the dominant users of ride-hailing services. As motorcycle taxi services are relatively cheaper and accessible, young and lower-income commuters, including informal workers, prefer ride-hailing services. The study also suggests the impact of the intensification of inter-peri-urban mobilities of post-suburbanisation on the increasing demand for ride-hailing. In this sense, the study proposes that the long-overdue strengthening of metropolitan-scale governance is urgently required. It is also crucial for metropolitan governments to issue appropriate inter-district public transportation policies and engage in development.

Introduction

The rapid development of urban areas has sprawled into their surrounding areas due to factors like migration, population growth, infrastructure expansion, and high land prices. Urban areas have transformed into metropolitan areas with a persistent tight relationship and connectivity between core and peripheral regions. This has increased the demand for transportation modes and infrastructure in metropolitan areas, leading to continuous expansion and diversification, such as highways in US metropolises, commuter trains in Europe, and bus rapid transit in developing country cities like Bogota (Colombia) and Jakarta (Indonesia).

In Indonesia, problems in the metropolitan transportation sector are considered severe. For instance, in the JMA, there are more than 1.3 million private cars, 2.6 million motorcycles, and 800,000 buses and trucks, coupled with heavy traffic in areas with limited road space (i.e., less than 10% of the surface, while internationally the applicable standard is 20–25%). Meanwhile, peripheral districts in the JMA, such as Bogor, Depok, Tangerang and Bekasi, have contributed to the high mobility rate within the metropolitan area. Jakarta's traffic is often

described as an 'urban nightmare' due to the absence of a good and reliable public transport system, leading to an increase in private vehicles (Steinberg, 2007). On the other hand, there is a relatively high share of motorcycles in Bandung and Jakarta metropolitan areas, 70% and 60%, respectively (BPS, 2018).

In developing countries, such as Southeast Asian countries, with poor public transportation, motorcycle taxis play an important role in mobility. Previous studies in the Jakarta metropolitan area suggested that commuters also use motorcycle taxis as an alternative to commuter train and bus rapid transit (Medeiros et al., 2018). Thus, with the growth of sharing platforms, fierce competition arises between conventional motorcycle taxis and ride-sourcing services (Irawan et al., 2020). In Indonesia, conflicts between online ride-hailing platforms and traditional ride-sharing drivers have led to bans on online drivers entering specific residential areas due to intense competition.¹

Presently, there is a lack of studies on the sharing economy, particularly ride-hailing, and its significance for commuters' mobility in Indonesian metropolitan areas. Existing studies on ride-hailing focus on the economic and legal implications of competition and predatory pricing (Azzuhri et al., 2018; Aggriawan, 2019), or how the emergence

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¹ <https://metro.tempo.co/read/692356/inilah-kawasan-anti-gojek-yang-wajib-diketahui-driver-gojek/full&view=ok>

of ride-hailing has led to an imagined community represented by the green colour in jackets and helmets, fostering solidarity, trust and safety for drivers (Natadja and Setyawan, 2016). Meanwhile, studies on metropolitan transportation primarily focus on commuter characteristics (Susantono et al., 2011), travel and mode choice (Ilahi et al., 2021; Liu et al., 2020; Rizki, Joewono, and Belgiawan, 2019), and willingness to pay (Belgiawan, Ilahi, and Axhausen, 2019). There is also a growing interest in metropolitan infrastructure governance, such as the important role of the state-owned company PT Jasamarga on toll road development (Firman, 2012), the issue of sources of clean water of the North Bandung Area (Kawasan Bandung Utara or KBU) (Argo, 2015) and the recent call to address the fragmented transportation planning problem (Miharja et al., 2021).

Recent studies have attempted to integrate the issues mentioned, revealing that ride-sourcing services support the existence of public transport in the JMA, as shown by Medeiros et al. (2018) and Irawan et al. (2020). However, there is a dearth of studies examining the importance of ride-sourcing for metropolitan commuters.

This study examines the importance of motorcycle-based ride-hailing as a commuting mode in the JMA and BMA. With a higher share of users and a growing debate on motorcycle-based ride-hailing (Anggriawan, 2019; Irawan et al., 2019; Senbil et al., 2006; Roberto, 2016), this study focuses on the importance of ride-hailing as a commuting mode in the JMA and BMA.

While the JMA stands as Indonesia's largest metropolitan and economic centre, the BMA is a relatively small metropolitan area, with Bandung recognized as a tourism destination and education hub. Both case studies are expected to shed light on the extent to which commuters' socio-economic and mobility characteristics determine ride-hailing use. They addressed several interrelated questions: What determinant factors foster or hinder ride-hailing among commuters? How does the spatial form of metropolitan areas affect the importance of ride-hailing for commuters? Finally, what policy implications may arise from the increasing importance of ride-hailing in metropolitan areas?

To address these questions, we employed quantitative descriptive statistics and logistic regressions using the Commuter Household Survey 2017 for BMA and 2019 for JMA, purchased from the Statistics Agency (Badan Pusat Statistik/BPS). The data included sample respondents from all 18 districts in Jakarta and Bandung metropolitan areas, representing 2459,990 and 403,063 commuters in JMA and BMA, respectively.² In these metropolitan areas, the population growth rate of the surrounding districts is significantly higher than that of the core cities (Firman, Kombaitan, and Pradono, 2007).

This article is structured as follows. First, we critically review the emergence of the sharing economy and expansion of metropolitan areas. Next, we elaborate on the present metropolitan area's present socio-economic, spatial and infrastructure characteristics. In the subsequent section, we present research data and methodologies. In the analysis section, we highlight the importance of population socio-economic and infrastructure characteristics in contributing to the expansion of metropolitan areas through ride-hailing. The final section concludes the discussion.

Literature framework

The phenomenon of the 'sharing economy' arises from technological developments that simplify 'sharing' activities, encompassing sharing goods and services in physical or non-physical form through information systems on the Internet. Also known as collaborative consumption, it involves peer-to-peer activities to access, provide, or share goods and services, coordinated through community-based online services (Hamari et al., 2015). Fundamentally, the sharing economy has

contrasted from the conventional economy assumption of scarcity, enabling people to collaboratively leverage underutilised inventory through cost-based sharing (Zervas, 2017). Leveraging technological advancement with information technology platforms, the sharing economy empowers users to share and take advantage of things that were not used before (Quattrone et al., 2016).

In this sense, collaborative consumption adheres to a lifestyle more concerned with 'sharing' than owning the goods or services. Consumers would have higher benefits because the costs incurred to consume an item or service will be cheaper, while waste will be reduced or avoided (Gutierrez et al., 2016). These peer-to-peer or sharing economy platforms have entered various sectors of the economy, such as the transportation, finance and accommodation (Quattrone et al., 2016). For instance, Uber has pioneered the sharing economy in transportation, Kickstarter in the financial sector (particularly crowdfunding), and Airbnb in accommodation. Here, we examine the cross-section between metropolitan areas and the sharing economy, focusing on commuters' mobility. Commuters perceive digital platforms as important tools for identifying needs and enabling more efficient use of transport modes, considering high levels of inequality. This may also alleviate traffic and guide policymakers to understand the most appropriate transport for each region (CIPPEC, 2018).

The seminal book by Sundarajan (2016) argues five characteristics of the sharing economy that disrupt the current economy. The implications of each characteristic could be applied to the ride-hailing applications. Firstly, the predominantly market-based economy has led to peer-to-peer exchange of goods and services, allowing limited space for government regulations. For instance, safety and price are negotiated between peers in a way that does not necessarily follow regulations. Secondly, high-impact capital, due to new opportunities for assets, skills, time and money for peers, is involved in the exchange of services. The sharing of resources between peers allows cost minimisation and, thus, increases revenue. Thirdly, crowd-based 'networks' rather than centralised institutions or 'hierarchies' allow further deployment of sharing resources between peers. Fourthly, there is a blurring line between the personal and the professional as the supplies of labour and services are often commercialised. Lastly, vagueness exists between fully employed and casual labour, independent and dependent employment, and work and leisure. These grey lines raise questions about the effect on the peers, whether they are ride-hailing users (Irawan et al., 2020) or the drivers (Fanggidae et al., 2016).

However, the impact of the arrival of such disruptors in markets and ecosystems remains a question. For instance, the literature suggests that socio-demographic characteristics, including age, gender, income and educational level, are key predictors of ride-sharing adoption. Ride-sharing consumers are typically young, well-educated and predominantly from middle-income groups (Dias et al., 2017; Rayle et al., 2016; Pongprasert and Kubota, 2017). They are also likely to undertake short trips (Tuffour and Appiagyei, 2014). There is also a wide body of literature that examines the regulations and institutions required to govern the new economy model (Ferreira et al., 2018) and the social conflicts that arise from this sharing economy model (Natadja and Setyawan, 2016).

Metropolitan areas and sharing economy

The phenomenon of metropolitan areas and megacities is the most attractive factor in urbanism (Kotkin, 2014). Megacities result from agglomeration in urban areas following the availability of amenities, socio-economic opportunities, and the benefits of agglomeration, which increases urban competitiveness (Kourtis and Nijkamp, 2013). As such, megacities are not only a result of large-scale urbanisation but are also a result of globalisation processes and competition. With the onset of the increasing connectivity between countries and the investments from globalisation, this leads to the issue of urban competitiveness of megacities.

² We attempted to use the Surabaya metropolitan area; however, the data shows very small number of samples that uses ride-hailing.

As such, in the recent decade, there has been a need for urban management to navigate urban growth as well as risk and dysfunction, either through market processes, education, planning, technology or infrastructure provision (Sorensen and Okata, 2011). As [Pojani \(2020\)](#) argued, the problem with transportation planning and development is implementing the acquired knowledge into practice to realise local urban transport sustainability in Indonesia, the Philippines and Thailand. The problems above have increased the importance of sustainability notions such as global climate change, technological advancement and governance change to reduce environmental footprints. Furthermore, the literature also suggests that the issue has become magnified in developing countries as infrastructures are neglected for a large share of the population with urban issues, including poverty, sickness and congestion.

However, there is a lack of communication between government regulations and the sharing economy. This may be a sign that there are no incentives to increase shared mobility market penetration or shared mobility can provide no clear benefits. Literature suggests several reasons; firstly, there is no clarity on the impacts on the sustainability of shared mobility. There are contrary implications between car-sharing and bike-sharing that may reduce CO₂ by limiting users to purchasing a private car, and ride-hailing (such as Uber, Lyft) may induce more trips ([Castellanos, Grant-Muller and Wright, 2021](#); [Santos, 2018](#)).

Secondly, the government and planners believe that shared-mobility services cannot replace public and mass rapid transport ([Di Bartolo et al., 2020](#)). For instance, a comparison study by [McKenzie \(2020\)](#) shows that ride-hailing could not escape traffic congestion that increases time travel, especially on weekends, suggesting micro-mobility services are faster in the city than ride-hailing automobiles in large cities.

Thirdly, there exists a knowledge gap between the government and planners regarding how the community engages with on-demand ride-sharing services. This bottom-up market planning shapes the future of urban environments. Presently, literature and policies have neglected the physical space and urban projects essential for constructing a shared identity and brand positioning of a sharing city ([Sánchez-Vergara, Ginieis, and Papaioannou, 2021](#)). The case in Australia, for example, questions how this market consumption in sharing cities may disrupt the provision of public transportation ([Curtis et al., 2019](#)). Studies have shown that governments should closely examine whether ride-hailing services may reduce vehicle use and ownership while attracting more customers ([Gerhrke, Felix, and Reardon, 2019](#); [Rayle et al., 2016](#)).

Jakarta and Bandung metropolitan areas

The Jakarta metropolitan area (JMA) is considered the second-largest metropolitan city in the world, with the fastest urbanisation rate in Southeast Asia. Jakarta city forms the core metropolitan area (highlighted in red), while surrounding districts constitute the peripheral regions (highlighted in green) ([Fig. 1](#)). There is a significant wage disparity among JMA districts, with average wages in bordering districts being less than 5 million IDR, whereas non-bordering districts' wages exceed 7 million IDR. Consequently, this disparity affects transportation modes, as nearly 60% of motorcycle users earn a minimum wage, and 73% of the population with a minimum wage own a motorcycle for commuting, while others rely on public transportation ([Aritenang, 2022](#)). The study also reveals that the poorest group, with a salary of less than 2 million IDR, is the highest group that uses public transportation, with motorcycle (51%), minibus (20%) and commuter train (17%). This preference for buses over ride-hailing for lower-income people was also observed by [Irawan et al. \(2020\)](#).

A recent study revealed that 63% of commuters from peripheral districts travel daily to Jakarta for work and school ([Aritenang, 2022](#)). Furthermore, a higher percentage of commuters in the lower-income group travel compared to those with higher income within Jakarta, accounting for 44% and 37%, respectively. On the other hand, the study also showed a higher share of higher-income group commuters

commuting between peripheral regions. These figures suggest Jakarta, as the core metropolitan area, serves as the employment centre and also hosts a significant population of lower-income individuals.

Various studies have examined the growing number of real-estate assets, gated communities and residential areas, and their impact on the economic and social segregation in the JMA ([Roitman and Recio, 2020](#); [Herlambang et al., 2019](#); [Ellisa, 2014](#)). The phenomenon's cause lies in the observable polarisation of middle- and upper-income residents, drawn to exclusive lifestyles, heightened security, first-class facilities and urban designs within new towns that limit interaction between the upper-middle class and lower-upper class ([Firman, 2004](#)). Examples include projects like Bukit Sentul, Lido Lake Resort in Bogor, Kota Legenda, and Lippo Cikarang in the Bekasi district, featuring diverse housing, industrial estates, office buildings, and other resort facilities ([Kenichiro, 2001](#)). As these new towns are dominated by relatively young commuters, this has led the metropolitan area to face various urban problems, including traffic congestion contributing to the increasing fiscal issues and losses.

Jakarta's total losses due to congestion reached Rp 67.5 trillion, according to the Jabodetabek Transportation Management Agency report (2017). Meanwhile, the accumulation of vehicles and congestion in Jakarta has led to an alarming level of pollution ([Steinberg, 2007](#)). To overcome this problem, the DKI Jakarta government has implemented several strategies on public transportation, including those on bus rapid transit (BRT) TransJakarta and commuter railway. Presently, the BRT system runs to the south (from University of Indonesia to Manggarai), east (Pondok Kelapa-Blok M) and the Jabodetabek Integration Corridor Busway (Jakarta-Tangerang, Jakarta-Bekasi and Jakarta-Depok/Bogor). To strengthen the BRT connectivity in the metropolitan area, feeder buses operate on the existing highways that run between Jakarta and peripheral districts, including Poris Plawad (Tangerang City), Bumi Serpong Damai (South Tangerang City), Depok Terminal (Depok City), Botani Park (Bogor City) and Bulak Kapal (Bekasi City).³

The Bandung metropolitan area has long been identified, with Bandung City being the core (red) and surrounding Bandung, West Bandung and Sumedang Regencies, as well as Cimahi City, as the peripherals (green) ([Fig. 2](#)). [Argo \(2015\)](#) found that Bandung City comprised more than 44% of the metropolitan area's GRDP, revealing the peripherals' high dependence on and connectivity to Bandung City as the metropolitan core. The Bandung metropolitan sprawling area develops with the expansion of built-up areas that follow the road system, especially the east-west axis. Particularly, the west axis is thicker than the east axis, indicating that the connection to Jakarta and Cimahi Cities is more intensive than the connection to Sumedang.

[Chapman and Prothero \(2012\)](#) found that a significant portion of commuters lived in Cimahi and worked in Bandung. A recent study by [Rustiadi et al. \(2020\)](#) suggested that better accessibility and a lack of land-use control capacity had led to the emergence of potential conurbation between Jakarta and Bandung metropolitan areas, including the districts in between: Cianjur, Karawang and Purwakarta Regencies. Nevertheless, there is also potential for further development towards the west with high-speed rail in South Bandung ([Hudalah et al., 2021](#)), population growth and housing development in Bandung Regency ([Vitriana, 2020](#)), and the presence of universities in Sumedang Regency ([Maulina and Rizal, 2017](#)).

Both metropolitan areas have developed spatially and economically, confirming the decline of urban primacy in core regions, high urban population growth, and strong linkages to the global economy, as predicted by [Firman, Kombaitan, and Pradono \(2007\)](#). The economic and spatial growth also intersects with another body of literature on metropolitan issues of social justice and inequality. In Indonesia findings about the gender link with ride-hailing use are inconclusive. [Belgiawan, Joewono, and Irawan \(2022\)](#) found no significant relationship between

³ See <https://transjakarta.co.id/peta-rute/>

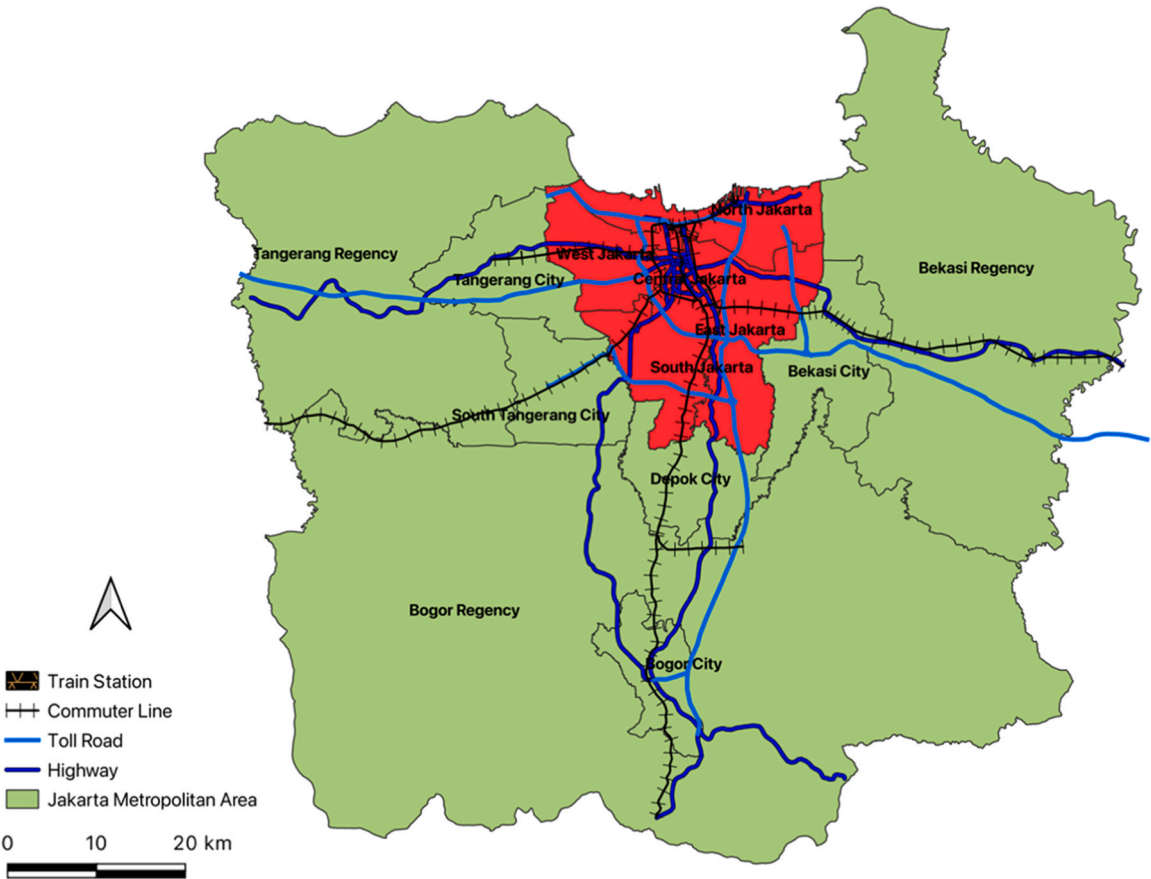


Fig. 1. The Jakarta Metropolitan Area (JMA).

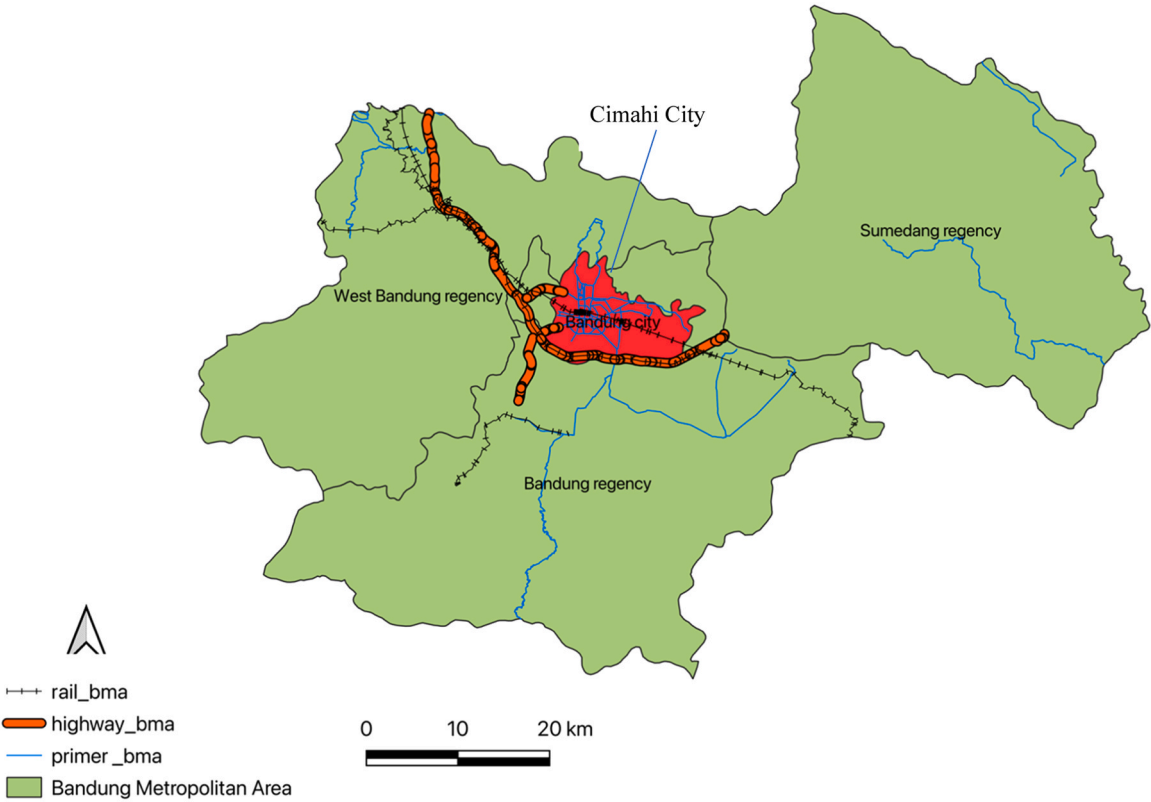


Fig. 2. The Bandung Metropolitan Area (BMA).

ride-sharing use in Bandung, while studies in Jakarta and Jogjakarta found a relationship between gender, education and income, either directly or indirectly through income, education, and service quality (Irawan et al., 2020b; Irawan et al., 2020a). On the other hand, there is evidence that gender differences influence BRT use, particularly among women, leading to initiatives aimed at increasing female ridership, such as offering female-only BRT carriages (Gaduh et al.; Witoelar et al., 2017).

The emergence of the sharing economy

Sharing economy platforms in Indonesia have grown rapidly in the last decade. Previous studies have highlighted that transportation policies, such as congestion charges and value capture, have rarely been adopted. Additionally, integrated physical infrastructure needs improvement; for instance, mass transit development was not accompanied by accessibility and maintenance of pedestrian routes that lead to stations (Gaduh, Gračner, and Rothenberg, 2022; Pojani, 2020). As such, ride-hailing provides alternative transportation modes, as motorcycle taxis are easier to locate and access, thereby allowing users to reach their destinations more efficiently, thus giving them shorter travel times (Irawan et al., 2020).

The platforms offer new services in addition to the main transportation services, such as food delivery and financial services (Wirawan and Oktivera, 2015). Specifically, in e-ride-hailing, key considerations for choosing an online motorcycle taxi for daily transportation needs are technical support and price transparency. The increasingly congested road conditions, especially during peak hours, make online motorcycle taxis an alternative. For instance, *Gojek* and *Grab*, both bike ride-sharing platform companies that initially connected riders and prospective passengers, have expanded in size to multisided platforms (MSPs).

The survey report by Afrina et al. (2017) showed that the main reasons for using motorcycle taxis are shorter travel time, avoiding traffic congestion and jostling in public transportation. This study also indicated that public interest in using public transportation is still substantial (65%), provided it is available in greater quantity and better service quality. The study revealed that price was not a priority in considering the choice of transportation type, and ride-hailing emerged as an alternative solution for users in dense and congested urban areas.

However, the emergence of e-ride hailing was problematic. Ride-hailing in Surabaya, for instance, revealed the rejection of Uber. The company was banned from public places such as airports, bus stations and railway stations. Although Uber drivers were allowed to take customers to these places, they were prohibited from picking up customers there. Nevertheless, customers continued to choose Uber as an airport transportation alternative due to its low price (Saadah, Yasmine, and Mubah, 2017). In another case, Depok City, a peripheral city in the JMA, became the first city to have regulations regarding online motorcycle taxis, i.e., Depok Mayor Regulation Number 11 of 2017 (Afrina et al., 2017). This regulation was a response to the increasing tensions between online motorcycle taxi drivers and drivers of other modes of transportation. The regulation stipulates that online motorcycle taxi company owners must provide special parking places for drivers, and those drivers are expected not to park or hang out on the side of the road, such as at bus stops, road shoulders or other public locations. The regulation also prohibits picking up customers in bus terminals and along roads which are travel routes of public transportation.

Data and methodology

This section discusses the data and methodology applied in the study, using commuter household data to illustrate the socio-economic and geographic dynamics in the metropolitan areas of Jakarta and Bandung.

Data

The Commuter Household Survey provided a general overview of transportation in Jakarta and Bandung metropolitans. The data included comprehensive information on commuters' residential location, economic and social characteristics (income, job details, educational background, and health conditions) and daily commuting activities (destination and transportation modes). The Commuter Household Survey benefits from two earlier statistic surveys (Sofiyandi and Siregar, 2020). First, he Indonesia's 2010 population census data to determine spatial stratified sampling strategy with a two-stage sampling selection method at the sub-district level, and second, the annual Indonesia National Labour Force Survey (*Survei Angkatan Kerja Nasional*/SAKERNAS) to determine the sample distribution.

The JMA commuter household survey data in 2019 covered thirteen districts, comprising eight municipalities and three regencies from three different provinces within the JMA: DKI Jakarta Province (i.e., Central Jakarta, North Jakarta, South Jakarta, East Jakarta and West Jakarta) as the core metropolitan area, and the West Java Province (i.e., Depok, Bogor, Bogor City, Bekasi and Bekasi City) and Banten Province (i.e., Tangerang, Tangerang City and South Tangerang) as the peripheral regions. This delineation of the JMA follows previous studies conducted elsewhere (Aritenang, 2021; Hudalah and Aritenang, 2017; Hudalah et al., 2013). The BMA commuter household survey data for 2017 included five districts: Bandung City (core metropolitan area), as well as Bandung, West Bandung, Sumedang Regencies, and Cimahi City (peripheral regions), following Vitriana (2020) and Miharja et al. (2021).

The data included more than 4000 respondents, representing more than 2.7 million commuters across the metropolitan areas. These data were aggregated at the district level in each metropolitan area, which allowed for descriptive and comparative analyses.

Research methodology

The following table presents the number of commuter respondents from the peripherals of the JMA and BMA, totalling 2856 (representing 2236,609 commuters) and 1112 (representing 580,921 commuters), respectively (Table 1). The table indicates that the largest share of peripheral commuters was from Bogor Regency, Depok and Bekasi Cities, constituting more than 56% of the commuters to the core metropolitan area. In the Bandung metropolitan area, more than 75% of peripheral commuters were from Bandung and West Bandung Regencies, suggesting these districts served as dormitory areas surrounding Bandung City.

The graphs below depict movement flows from peripherals to core metropolitan areas. In the Jakarta metropolitan area, the graph shows that the number of commuters within the core metropolitan area (DKI Jakarta Province) was higher than the number of commuters from each peripheral district (Fig. 3A). On the other hand, only a small fraction of people from the core metropolitan area commuted to the peripheral districts. The graph also reveals that the largest share of commuters to the core metropolitan area was from Depok City, Bekasi City and Bogor

Table 1
The number of Commuters to Core Metropolitans, BMA 2017 and JMA 2019.

Peripheral Districts	Core Metropolitan	Number of Commuters
Bogor Regency	Jakarta Province	483,053
Bekasi Regency		225,379
Bogor City		86,347
Bekasi City		374,188
Depok city		405,945
Tangerang Regency	Bandung city	227,937
Tangerang City		236,358
South Tangerang City		197,402
Bandung Regency		301,252
Sumedang Regency		54,459
West Bandung Regency		141,698
Cimahi City		83,512

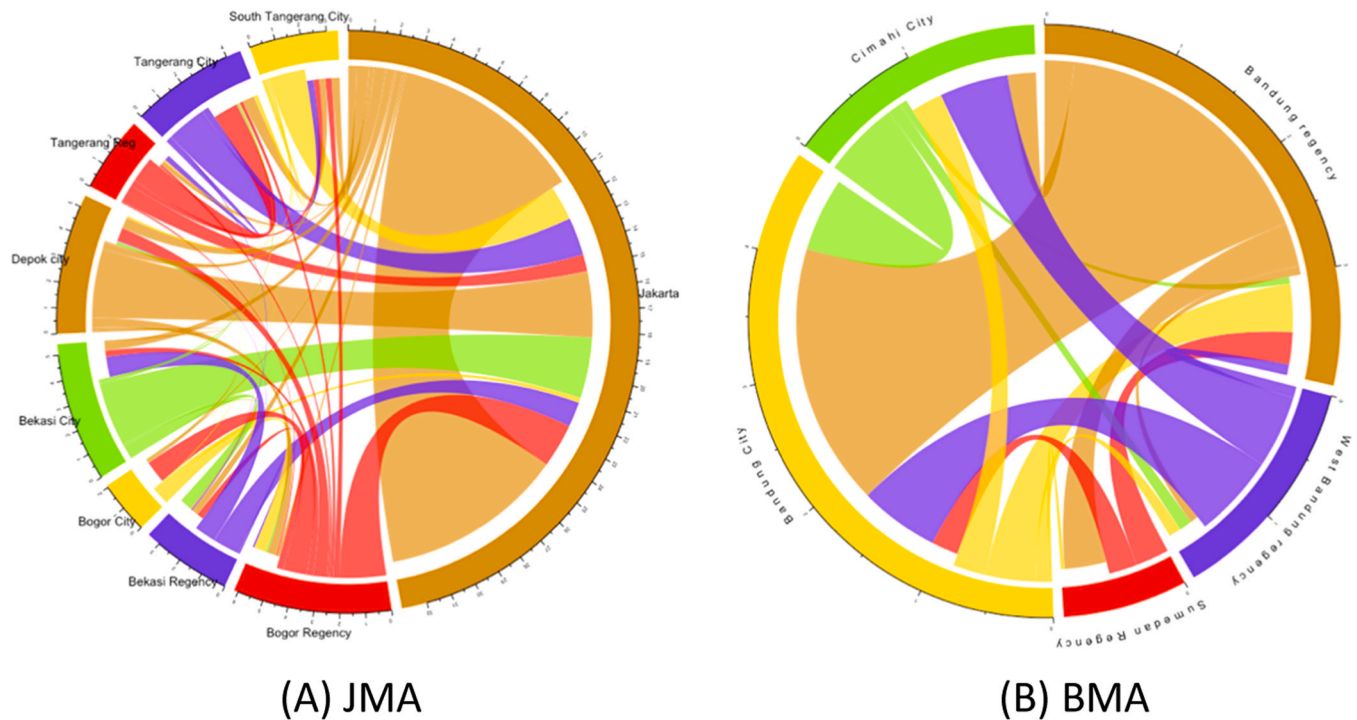


Fig. 3. The JMA (A) and BMA (B) Commuter Flows, 2017 and 2019.

Regency, implying a significantly larger share of commuters from West Java Province districts than those from Banten Province districts. Interestingly, the graph also indicates strong connectivity between neighbouring districts, such as between Bekasi City and Regency and between Tangerang City and Regency.

The second graph presents the commuting mobility in the Bandung metropolitan area (Fig. 3B). The largest share of commuters to the core metropolitan area (Bandung City) was from Bandung Regency, followed by commuters from West Bandung Regency and Cimahi City. Contrarily, the graph further shows that most commuters were from the core metropolitan area towards Bandung Regency and Cimahi City. Furthermore, we found a large share of commuters from West Bandung Regency to Cimahi City, considering its proximity to Bandung City.

Overall, the graphs suggest the process of post-suburbanisation with the diversity of commuting destinations in the JMA and BMA, featuring a higher proportion of peripheral commuters traveling to other peripheral districts within the metropolitan areas.

This paper conducted two analyses. First, we analysed the characteristics of motorcycle-based ride-hailing users using descriptive analysis to investigate commuters' characteristics and commuting behaviour in both JMA and BMA. Second, logistic regressions were performed to examine determinants of the probability of using motorcycle-based ride-hailing in both metropolitan areas. The dependent variable is the categorical variable indicating ride-hailing use (0/1), and independent variables include three main aspects: socio-economic conditions (age, sex, type of activities, salary, job type), commuting pattern (commuting distance, time of commuting) and destination (from or to core/peripheral regions).

Ride-hailing commuters' characteristics and proportion in the metropolitan areas

This section examines the characteristics of commuters who used ride-hailing and their proportion to overall commuters in the Jakarta and Bandung metropolitan areas.

First, we analysed the importance of ride-hailing by examining the proportion of commuters using the mode in the JMA and BMA (Table 2).

Table 2
Proportion of Commuters to Core Metropolitans, BMA 2017 and JMA 2019.

	Jakarta Metropolitan Area (JMA) (%)	Bandung Metropolitan Area (BMA) (%)
Motorcycle-based		
Ride Hailing	4.62	3.27
Wage		
< 2 mio IDR	4.54	4.03
2–2.9 mio IDR	3.78	3.54
3–3.9 mio IDR	2.54	0.28
4–4.9 mio IDR	4.71	3.73
Status		
Informal	11.74	
Labour/Employee	4.05	2.48

The share of ride-hailing commuters in the JMA was slightly higher than that in the BMA, at 4.62% and 3.27%, respectively. We also examined the minimum wage in each metropolitan area. In the JMA, the minimum wage was Rp4.7 million, indicating that commuters earning minimum wages tended to use ride-hailing more than lower-wage earners. Meanwhile, In the BMA, with a minimum wage of Rp3.7 million, the proportion of commuters declined with increasing wages, suggesting that higher wage earners preferred other transportation modes, such as private motorcycles or public transportation, for commuting. However, the share of commuters who travelled with ride-hailing was much higher among those earning Rp4–4.9 million in both the JMA and BMA, indicating other factors attracting them to use this mode, such as higher reliability than public transportation or a better ability to confront traffic congestion.

The employment status revealed interesting findings, with more than 11.7% of informal workers using ride-hailing in the JMA, a figure much higher than the 4% share among workers/employees. This suggests that income certainty allows employees to consider other transportation options, such as private or company vehicles. In the BMA, the figure was smaller, with only 2.45% of commuters from the peripherals using ride-

hailing. However, the survey did not provide data on the number of commuters working in the informal sector in the BMA. The table further reveals that only 5% and 3.46% of commuters from peripherals used ride-hailing to travel to core metropolitan areas in the JMA and BMA, respectively.

Second, we analysed the socio-economic conditions and spatial flow of motorcycle ride-hailing commuters (Table 3). In the JMA, commuters using motorcycle ride-hailing were predominantly in the working age range of 23–40 years old, followed by young adults and teenagers. Meanwhile, in the BMA, motorcycle ride-hailing commuters were mainly young adults and teenagers (below 23 years old), constituting more than 43 percent of total commuters using this service. Furthermore, in both metropolitan areas, senior adults aged 40–60 and those above 60 were the least represented among commuters using motorcycle ride-hailing.

In both metropolitan areas, the number of female commuters significantly exceeded male commuters. This suggests an imbalance in gender proportion in metropolitan areas. For commuting purposes, work was the primary activity in both areas (approximately 51–56%), followed by students commuting to school (43–48%). Ride-hailing was preferred for its speed, accessibility, and convenience, confirming the reliability of motorcycle ride-hailing amid congestion, as indicated in previous studies (Irawan et al., 2019; Sunitiyoso et al., 2022). Interestingly, safety and cost were not the primary concerns for ride-hailing

Table 3

Proportion of Ride-hailing Commuters to Core Metropolitan by Socio-economic and Spatial Flow, BMA 2017 and JMA 2019.

	Jakarta Metropolitan Area (JMA)	Bandung Metropolitan Area (BMA)
Age		
<18 yo	37.19	25.48
18–23 yo	17.62	38.02
23–40 yo	24.47	13.74
40–60 yo	20.71	22.75
>60 yo	0.01	0.01
Sex		
Male	40.88	29.83
Female	59.12	70.17
Activities		
Work	51.21	56.41
School	48.79	43.69
Informal Education/Courses	0.01	0.01
Reason		
Faster	59.91	31.68
Convenient	19.79	5.14
Safety	0.00	1.57
Cost	1.59	1.84
Easy to access/Practicality	18.7	59.76
Others	0.01	0.01
Wage		
Below UMR	27.34	56.82
Above UMR	72.66	43.18
Status		
Informal	5.86	-
Labour/Employee	75.24	59.79
Self Employed/Entrepreneur	18.9	40.21
Travel Distance		
<10 km	52.36	55.66
10–19 km	20.11	33.12
>20 km	27.54	11.21
Travel Time		
<30 minute	66.38	47.21
30–60 minute	22.76	41.23
>60 minute	10.86	11.56
Destination		
Peripherals to core	2.65	3.87
Peripherals to peripherals	2.67	2.80

users.

Next, the wage figure range suggested that the commuters who use motorcycle ride-hailing in the BMA are dominated by commuters with wages below the regional minimum wage (*Upah Minimum Regional/UMR*), contrary to Jakarta where the figures are 56.8% and 27.3%, respectively. Nevertheless, it should be noted the wide gap in UMR thresholds between the two metropolitan areas—4.9 million IDR in JMA and 2.8 million IDR in BMA. This may suggest other factors attracting the use of motor-based ride-hailing between metropolitan areas, such as higher reliability compared to public transportation (Irawan et al., 2019) or a better ability to confront traffic congestion to some extent (Aritenang, 2022).

The employment status revealed that employees or workers were likelier to use motorcycle ride-hailing to commute compared self-employed workers. The employment status revealed that only 4.34% of informal workers use ride-hailing in the JMA. The proportion of employees or workers using motorcycle ride-hailing is significantly higher in the JMA than in the BMA, being 75% and 59.8%, respectively. In addition, despite the lack of data on the sample of informal workers, we could still identify a small proportion of informal workers as users of motorcycle ride-hailing. It to be noted that the survey did not have samples of commuters that work in the informal sector in BMA.

This finding confirms the distinct characteristics of both metropolitan areas: the JMA is dominated by offices and business premises, while Bandung City, the core of the BMA, is renowned for its schools and universities.

We also examined the spatial mobility of motorcycle ride-hailing. The distance figures suggest that the proportion of short-distance commuting is much lower in the JMA, with commuting distances of less than 20 km accounting for more than 72%. In contrast, short-distance commuting makes up 89% in the BMA. In the JMA, over 27% of commuters using motorcycle ride-hailing are on long-distance travels (> 20 km) daily. Furthermore, the share of commuters that travel less than one hour is similar between the metropolitan areas, about 89%. However, the data reveals that the share of commuters in the JMA using ride-hailing for less than 30 minutes is much higher than in the BMA, 66.3% and 47%, respectively.

A comparison was made in the commuter destination between the core and peripherals. The figures reveal that the commuter flow in the JMA was more diverse, with 1218,944 commuters (less than 59%) from peripheral areas commuting to Jakarta, with 33,233 (2.65%) commuters using motorcycle-based ride-hailing. Meanwhile, 350,055 commuters (68.12%) from peripherals commute to Bandung city, with 13,549 (3.87%) commuters using motorcycle-based ride-hailing.

Determinants of using motorcycle-based ride hailing

The following table presents the descriptive statistics of variables used in the logistic regression (Table 4).

Our analysis examines the determinants of odds ratio using ride-hailing in both metropolitan areas (Table 5). The socio-economic variables suggest similar determinants of using ride-hailing; the odds ratio of using ride-hailing are lower with older commuters, with the probability of using ride-hailing reduced to 0.98 as a person's age increase by 1 year. The odds of using ride-hailing are significantly higher among women, with 10.1 and 12.4 higher probabilities in Jakarta and Bandung metropolitan areas, respectively. However, in Jakarta metropolitan, the odds of using ride-hailing are high for commuters working in the formal sector and those with a higher salary. In Bandung, formal workers tend to have lower odds of using ride-hailing, and salary is an insignificant variable.

Furthermore, commuting patterns reveal a slight difference in ride-hailing users in JMA and BMA. In JMA, the odds ratio to commute using ride-hailing is higher (0.924) compared to BMA (0.454) for commuting distances less than 10 km. The odds ratio for using ride-hailing is lower for early-hour (before 7 am) commuters in JMA. In

Table 4
Descriptive Statistics, BMA 2017 and JMA 2019 BMA, 2017.

	Mean	SD	Min	Max
Mode (dummy)			0	1
Age (year)	31.49	13.69	6	85
Gender (M/F) dummy			1	2
Job formal/non-formal (dummy)			0	1
Salary (IDR mio)	3536,210	3176,461	0	50,000,000
Commuting distance (km) (range dummy)			1	7
Commuting before 7 am (dummy)			0	1
Destination (dummy)			0	1
Cost	11,592	10190	0	100,000
JMA, 2019				
	Mean	SD	Min	Max
Mode (dummy)			0	1
Age (year)	37.63	11.58	17	83
Gender (M/F) dummy			1	2
Job formal/non-formal (dummy)			0	1
Salary (IDR mio)	5518,431	5080,476	300,000	100,000,000
Commuting distance (km)	21,02	15,34	0	160
Commuting before 7 am (dummy)			0	1
Destination (dummy)			0	1
Cost	127,761	143,168.4	0	2500,000

BMA, the odds ratio is also significantly lower for early-hour commuters after 8 am. Lastly, destination variables suggest similar findings with commuting to core metropolitan areas and a higher ratio of travel cost to household expenditure, indicating a higher odds ratio for using ride-hailing. This implies commuters from the lower- and middle-income groups have a higher probability as consumers.

Discussion

This section provides an elaborative discussion of the findings and analyses presented in the previous section.

High socio-economic connectivity, poor infrastructure accessibility

Jakarta's transportation problems have been widely discussed in previous studies. Afif (2002) investigated the lengthy process of attempting to solve Jakarta's transportation problems across various studies, which are yet to provide any comprehensive solutions. For instance, the paper argued that the studies did not understand the 'bus priority' concept, viewing bus priority projects as temporary solutions. This led to reluctance in proposing significant changes to the existing

bus system and a focus on developing a mass rapid transportation (MRT) system.

Metropolitan development in Indonesia has been closely linked to the suburbanisation debate, extensively explored in prior studies, particularly on the JMA (Hudalah and Firman, 2012; Hudalah et al., 2013; LeGates and Hudalah, 2014; Firman and Fahmi, 2017; Aritenang, 2023). Limited research focuses on the BMA, though some studies recognise the metropolitan area to some extent (Webster, Cai, and Muller, 2014; Hudalah, Talitha, and Lestari, forthcoming). Our study confirmed the above literature, as shown by a strong connectivity between peripheral regions in the JMA and the persistent high dependence on the core metropolitan area in the BMA.

Social justice, metropolitan areas and the sharing economy

Our study confirmed the irony of ride-hailing among commuters. The main reason for preferring ride-hailing was its speed and practicality, also found by previous studies (Afrina et al., 2017; Irawan et al., 2020). Meanwhile, the urban van (angkutan kota/angkot) was banned. Ride-hailing became more tolerated and even emerged as an alternative transportation mode despite it being regarded as an informal mode of transportation (Medeiros et al., 2018; Syabri et al., 2013). Additionally, ride-hailing is more convenient due to the availability of alternative electronic payment, the presence of a complaint mechanism and discounted prices (Afrina et al., 2017). Our study found that convenience and practicality remained the main reasons for using ride-hailing, despite the government's pledge to enhance accessibility and reduce commuting time, which was to be achieved by 2010 (SITRAM, 2004). Interestingly, these reasons were not considered the most important in neighbouring countries such as Malaysia, where avoiding driving while under alcohol and parking prices were the most important. In Southeast Asia countries, price, safety and convenience were the main determinants for using ride-hailing (Chalermpong et al., 2022).

Our study also touched on another body of literature on the metropolitan issues of social justice and inequality. In both metropolitan areas, commuters from the lower- and middle-income groups have a higher probability of using ride-hailing. More than 82% of commuters spent less than Rp 200,000 for ride-hailing, both conventional and on-line. Given that this was less than 6% of their monthly expenditure, it is no wonder that the mode became an alternative commuter mode in the JMA and BMA. This advantage made ride-hailing the primary alternative transportation mode for lower-income commuters, informal workers and employees in both the JMA and BMA. In the JMA and BMA, the share of the lowest-income group was 29% and 67%, respectively. Nevertheless, this underscores the problem identified in Metro Boston, suggesting that the government should be concerned with the quality of public transportation if lower-income households are willing to pay for

Table 5
Determinants of Using motorcycle-based Ride-hailing Commuters, BMA 2017 and JMA 2019.

	Jakarta			Bandung		
	Coefficient	Odds Ratio	Std Error	Coefficient	Odds Ratio	Std Error
Socio-Economic						
Age (year)	-0.020	0.980***	(0.00)	-0.024	0.977***	(0.00)
Gender (M/F)	2.314	10.118***	(0.09)	2.518	12.400***	(0.31)
Job Formal/Informal	1.611	5.006***	(0.17)	-0.854	0.426***	(0.01)
Salary (IDR Mio)	0.316	1.357***	(0.01)	0.021	1.022	(0.02)
Commuting Pattern						
Distance (KM)	-0.079	0.924***	(0.00)	-0.790	0.454***	(0.01)
Commute before 7 am	-0.496	0.609***	(0.01)	0.038	1.039	(0.02)
Destination						
Commuting to core	0.949	2.582***	(0.03)	0.581	1.787***	(0.04)
Travel cost ratio to Expenditure	1.464	4.325***	(0.02)	1.442	4.231***	(0.06)
const	-26.580	-	-	-19.410	-	-
No. of Observation		2.459.990			403.063	
Adj R2		0.323			0.309	

* p<0.05; ** p<0.01; *** p<0.001.

short trips (MAPC, 2018).

As such, commuters have neglected safety and cost, suggesting that online motorcycle taxis are an alternative solution for urban areas, and they do not object to paying more for less time travel. Particularly, in JMA, the odds ratio to commute using ride-hailing is higher than BMA for shorter commuting distance, suggesting the lack of active or decent public transportation in the metropolitan area.

Related to gender, we found a significant number of female commuters using ride-hailing. Limited studies have examined the gender aspect of commuters in metropolitan areas. However, research on gender-based ride-hailing indicates a higher proportion of female commuters in cities such as Tehran (Etminani-Ghasrodashti and Hamidi, 2019) and across Southeast Asia (Chalermping et al., 2022; Lim and Fernandez, 2022). Furthermore, IFC (2020) highlighted several reasons for the high share of female riders. For example, ride-hailing enhances women's mobility and independence, with lower costs being crucial as they consider disposable income. In Indonesia, Irfan and Nooraeni (2019) shed light on the characteristics of female commuters. The 25–44 age group is dominated by female commuters at 56.55%, followed by the 15–24 age group at 25.18%. The percentage of female workers commuting for over 90 minutes was low at 28%. These figures may explain the large proportion of female ride-hailing users, as female commuters were mainly young and had relatively short commuting times.

Conclusion

This paper contributes to the current literature on metropolitan studies and travel behaviour concerning the ride-hailing services. The study identifies the characteristics of commuters who use ride-hailing and the importance of the transportation mode in the JMA and BMA. Our study was conducted with sample data on commuter respondents in the JMA and BMA. To the best knowledge of the authors, this is the first research to examine commuter data, revealing the critical role of the ride-hailing service in Indonesian metropolitan areas.

Our results indicate specific characteristics of ride-hailing service users among commuters in both metropolitan areas. Analyses report that young and lower-income commuters were the dominant users of the ride-hailing service. Within the sharing-economy framework, motorcycle taxis, being relatively cheaper and accessible (either through conventional informal pools/ojek or online), were preferred by young and lower-income commuters, including informal workers. However, the preference for ride-hailing services increased in both metropolitan areas once the salary exceeded the minimum wage, despite a decrease in the share of ride-hailing services by income. This confirms our other findings on the importance of ride-hailing services for work activities. This may be explained by the reliability of ride-hailing services in confronting prolonged and fatigued traffic congestion in metropolitan areas, especially for high-income earners who are mobile. Furthermore, our study also revealed a gender bias, where more than 72% of commuters who used ride-hailing services were females in both metropolitan areas.

The study also suggested the impact of metropolitan form on travel behaviour. The JMA has a larger size and a lower proportion of commuters traveling within the metropolitan area, while the BMA has a relatively smaller size and a higher percentage of commuters heading to the core metropolitan area. Our study showed the process of post-suburbanisation with strong connectivity between the peripherals in the JMA, such as between Bekasi City and Regency, and between Tangerang City and Regency. On the other hand, Bandung City remained the main destination for commuters from the peripheral districts. Furthermore, the presence of highways and alternative roads between the districts might also contribute to high connectivity between the peripheral districts in the JMA. The analysis also exposed shorter distances and durations for ride-hailing services among BMA commuters. In contrast, ride-hailing commuters in the JMA averaged more than one hour of travel covering 20 km daily.

This paper makes several contributions to the literature and policy implications. Firstly, it empirically enhances the urban regime literature by emphasising the intensified commuting during the post-suburbanisation process in the JMA. Over 40% of peripheral commuters now travel to other peripheral districts within the metropolitan area. On the other hand, the BMA case reveals a strong reliance of peripheral districts on the core metropolitan area, with less than 25% of peripheral commuters travelling to other peripheral districts. Additionally, the study contributes to the metropolitan studies literature in the global south, providing evidence that long-distance and -hour ride-hailing intensifies traffic congestion, especially in cities lacking public transportation modes like Jakarta and Bandung. The study also adds to the current sharing economy by highlighting the importance of ride-sourcing as an alternative transportation mode for public transport, particularly for lower- and middle-income group commuters.

Secondly, the study proposes several policy implications. (i) Considering the importance of ride-hailing as a transportation mode, there is a need for government policies to ensure safety and affordability for commuters who use the service on a daily basis, along with legal frameworks to regulate inter-district online ride-hailing. Additionally, (ii) evidence of post-suburbanisation in the JMA, with a high share of inter-peri-urban mobilities, highlights the increasing importance of safety and affordability policies for ride-hailing users. Improved accessibility and convenience in public transportation are also crucial for commuters. This may also apply to other global metropolitan areas that are experiencing post-suburbanisation following the intensification of inter-peri-urban mobilities. Ride-hailing may provide an accessible and convenient alternative to public transportation for commuters.

(iii) Thus, the long-overdue strengthening of metropolitan-scale governance is urgently required. This study confirms the presence of a diversity of mobility of commuters between peripheral districts, not solely towards core metropolitan areas, indicating an ongoing process of post-suburbanisation in both the JMA and BMA, albeit at different levels. Furthermore, metropolitan governance should enhance the provision of secure, convenient and cost-effective public transportation facilities. In particular, there is a need for stronger metropolitan-level authority and partnerships with the private sector, especially in the sharing economy sector, to effectively coordinate planning collaboration.

Declaration of Competing Interest

None.

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